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Wing Shape Variation in the Taxonomic Recognition of Species of *Diachlorus* Osten-Sacken (Diptera: Tabanidae) from Colombia

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Abstract

We evaluated the directional asymmetry between right and left wings and quantified the intraspecific and interspecific variation of the wing shape of 601 specimens of the genus *Diachlorus* to determine to what extent the geometrical variation discriminates six species distributed in six protected areas of Colombia. Geometric analyses were performed, integrating Procrustes methods, principal component analyses, cluster analyses, linear and quadratic discriminant analyses, and evaluations of shape changes. In *Diachlorus*, left and right wings did not present significant asymmetry but a geometrical analysis was allowed for species identification and, in some cases, the origin of the specimens using the variation of wing shape; the best-assigned species was *Diachlorus leticia* Wilkerson & Fairchild, while the worst was *Diachlorus jobbinsi* Fairchild, which also had the highest intraspecific variation, while *Diachlorus fuscistigma* Lutz had the lowest variation. *Diachlorus fuscistigma* and *Diachlorus leucotibialis* Wilkerson & Fairchild were the most similar species, while *D. leucotibialis* and *Diachlorus nuneztovari* Fairchild & Ortiz were the most disimilar. The specimens with the most different wing shape belonged to Chocó (especially those of *D. jobbinsi*), the geographically farthest area from the others in the study; however, no correlation was observed between geometric and geographical distances. Linear discriminants were better than nonlinear (quadratic) discriminant analyses in predicting species membership, but the opposite was true for predicting area membership. Based on our data, we hypothesized that other species of *Diachlorus* could also be discriminated using geometric morphometry of the wing shape.

Introduction

Tabanidae (Diptera: Brachycera) is a monophyletic family representing nearly 4400 described species (Wiegmann *et al* 2000, Morita 2008, Roskov *et al* 2014), with 27% (1205 species) distributed in the Neotropical region (Coscarón & Papavero 2009b, Henriques *et al* 2012). Nowadays, there are 234 species of Tabanidae recorded in Colombia (Coscarón & Papavero 2009b, Henriques *et al* 2012). Despite the increasing number of studies performed during the last 10 years in Brazil (Henriques 2006, Krolow & Henriques 2009, 2010, Limeira-de-Oliveira *et al* 2009, Turcatel *et al* 2010, Rafael *et al* 2012, Henriques & Krolow 2013, Gorayeb *et al* 2013, Gorayeb 2014, Krolow *et al* 2015), Chile (González 2004, 2006, 2009, 2014), and Ecuador (Buestán *et al* 2007, Cárdenas *et al* 2009), taxonomic research on Neotropical Tabanidae is still incipient.

The taxonomic classification scheme used for Tabanidae includes four subfamilies (Chrysopsinae, Pangoniinae, Scepsidinae, and Tabaninae), with further division into tribes (Mackerras 1954, 1955, Coscarón & Philip 1979). This classification and the principal taxonomical keys are based on

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external and genital morphological information of adults, and external morphology of known immature stages (Coscarón & Papavero 1993, 2009a, 2014).

The genus *Diachlorus* Osten-Sacken (Tabaninae: Diachlorini) is a group of small flies with yellow and black coloration; its pleura has a patch of pearly pollinosity; the wings have dark patterns of spots, mainly at the apex, and the eyes have characteristic patterns of color lines very similar to those presented in the genus *Chrysops* Meigen (Chrysopsinae: Chrysopsini) (Fairchild 1969). Fairchild (1972) and Wilkerson & Fairchild (1982) presented the taxonomical keys for this genus. Henriques & Rafael (1999) described two species, reaching 29 recognized species in the world (Coscarón & Papavero 2009b). The distribution of species of *Diachlorus* ranges from the southeastern USA and the Bahamas to Argentina, but they are absent from Chile and the Caribbees (Fairchild 1969, Coscarón & Papavero 2009a). *Diachlorus anduzei* Stone, *Diachlorus curvipes* (Fabricius), *Diachlorus fuscistigma* Lutz, *Diachlorus jobbinsi* Fairchild, *Diachlorus leticia* Wilkerson & Fairchild, *Diachlorus pechumani* Fairchild, and *Diachlorus xynus* Fairchild are distributed in Colombia, mainly in the Amazon region (Wilkerson & Fairchild 1982, Coscarón & Papavero 2009b).

Morphological interspecific variation has not been quantified in Tabanidae, although Cárdenas *et al* (2013), using classical morphometry, measured the body size, thorax volume, wing area, and wing loading in specimens of *Stypommisa* sp. (Tabaninae: Diachlorini), along an altitudinal gradient to evaluate the correlation of those variables with flying activity.

Right and left sides of the body can play different roles in taxonomic, evolutionary, and ecological understanding (Klingenberg *et al* 1998, Runyon & Hurley 2004, Savriama & Klingenberg 2011), therefore, the quantification of symmetrical differences can show whether there are different histories (left versus right side) in morphological variation, adaptation, or development of traits and thus define if the differential use of structures on the bilateral symmetry body plan is necessary (Swaddle 1997, Tomkins & Kotiaho 2001, Adams *et al* 2013).

Taxa, in general, exhibit different levels of variation. *Diachlorus pechumani* presents an intraspecific variation that generates some taxonomic recognition problems, and it has been divided into two subspecies (Wilkerson & Fairchild 1982). Also, some species of the genus (e.g., *D. fuscistigma*, *D. jobbinsi*) present an intraspecific variation in their external morphology, which, in some cases, makes the taxonomic identification complicated (Wilkerson & Fairchild 1982). In Diachlorini genera, as in other tribes, the intraspecific variation is remarkable. Species of the genus *Chlorotabanus* Lutz, for example, show high regional variation (Krolow & Henriques 2010). The genus *Dichelacera* Macquart also presents high morphological variation, while some of their species are very similar to each other, e.g., *Dichelacera costaricana* (Fairchild) and *Dichelacera apicalis* (Fairchild) (Fairchild & Philip 1960, Wilkerson 1981, Henriques & Rafael 1993).

Many approaches can be used to evaluate the morphological variation, and geometric morphometric analysis became an efficient tool to register shape changes that could not be easily detected with the naked eye, and therefore, is useful for taxonomical, ecological, and phylogenetic studies (Rohlf 1990, Rohlf & Marcus 1993, Weaver *et al* 2014). As insect wings are almost transparent and flat and have many comparable regions, these are used to avoid the subjectivity on geometric data capture (Lorenz & Suesdek 2013).

In the present study, we evaluated the directional asymmetry between right and left wings and quantified the intraspecific and interspecific variation of the wing shape for determining to what extent the geometrical variation discriminates the species of the genus *Diachlorus* from Colombia.

Material and Methods

Species

The biological material examined corresponded to females of six species: four previously known (*D. curvipes*, *D. fuscistigma*, *D. jobbinsi*, and *D. leticia*) and two unknown species from Colombia (*Diachlorus leucotibialis* Wilkerson & Fairchild, from Amazonas, Caquetá, Putumayo, and Vaupés, and *Diachlorus nuneztovari* Fairchild & Ortiz, from Amazonas and Putumayo).

Although there are other *Diachlorus* species in Colombia, they were not included because of the few specimens available (lower than 10 specimens for *D. pechumani* and *D. xynus* and none for *D. anduzei*). This research only used females because the main source of specimens was Malaise trapping and females of Tabanidae (most of them blood feeding) are easier to collect using these traps and are more abundant than the nectar-feeding males (Coscarón & Papavero 2009a, Lessard *et al* 2013).

Localities

The specimens included in this study belong to two collections of Tabanidae: "Instituto de Investigación de Recursos Biológicos Alexander von Humboldt (IavH)," collected in 23 protected areas of Colombia during the project "Diversidad de Insectos en Colombia" (Campos & Fernández 2002), and "Instituto de Ciencias Naturales (ICN-UNAL-Bogotá)," collected from two protected areas of the south of Colombia.

Six protected areas had the minimum number ($n>29$) of specimens defined (see the "Material and Methods" section) (Table 1). The "Parque Nacional Natural (PNN)

Table 1 Species and number of specimens by area (locality) used in this study.

	AMA	CAQ	CHO	MET	PUT	VAU	Total
Diachlorus curvipes	30		32	32	32		126
Diachlorus fuscistigma	32	32		32	32	35	163
Diachlorus jobbinsi	32	32	32		30		126
Diachlorus leucotibialis[a]	32	30			32		94
Diachlorus leticia						30	30
Diachlorus nuneztovari[a]	30				32		62
Total	186	94	64	64	190	65	601

AMA Amazonas, "PNN Amacayacu"; *CAQ* Caquetá, "PNN Serranía de Chiribiquete"; *CHO* Chocó, "PNN Utría"; *MET* Meta, "PNN Sierra de la Macarena"; *PUT* Putumayo, "PNN La Paya"; *VAU* Vaupés, "EB Mosiro Itájura Caparú."

[a] Species not previously registered in Colombia.

Amacayacu" (AMA), located in the department of Amazonas (3°17′28″S, 70°9′21″W), is a tropical rainforest ecosystem with flooded forests and an average temperature of 27°C; the "PNN Serranía de Chiribiquete" (CAQ), between the departments of Caquetá and Guaviare (0°43′0″N, 72°43′59″W), is composed of large rock formations surrounded by dense rainforests with an average temperature of 24°C; the "PNN Utría" (CHO) is in Chocó (6°1′14″N, 77°14′2″W) and is a cove in the Pacific Ocean, covered by tropical rainforests with an average temperature of 27°C; the "PNN Sierra de la Macarena" (MET), located in the department of Meta (2°36′32″N, 73°33′44″W), is composed of rainforests and Amazon savannas with an average temperature of 27°C; the "PNN La Paya" (PUT), in Putumayo (0°13′8″N, 75°14′46″W), corresponds principally to tropical rainforest with an average temperature of 26°C; finally, the "Estación Biológica (EB) Mosiro Itajura-Caparú" (VAU), in the department of Vaupés (1°04′0″S, 69°30′34″W), is on the border of Brazil, and its ecosystem is the Amazon forest with an average temperature of 27°C ["Sistema de Parques Nacionales Naturales de Colombia" and "Sistema Nacional de Áreas protegidas de Colombia (SINAP)"; http://www.parquesnacionales.gov.co/, accessed 27 January 2015].

Data collection

Specimens were previously determined using the taxonomic keys by Fairchild (1972), Wilkerson (1979), Wilkerson & Fairchild (1982), and Coscarón & Papavero (2009a). In order to gather geometric shape data to evaluate directional asymmetry in wing shape, subsamples were randomly taken (30 specimens for each species) from a sample of 601 total specimens (Table 1), taking photographs of left and right wings in a dorsal view. For further analyses, photographs were also taken of the right wing in a dorsal view for each one of the 601 specimens. For the photographs, the wings were placed over microscope slides without detaching them from the body. After that, photographs were taken using a stereoscope attached to a camera of 10.1 megapixels.

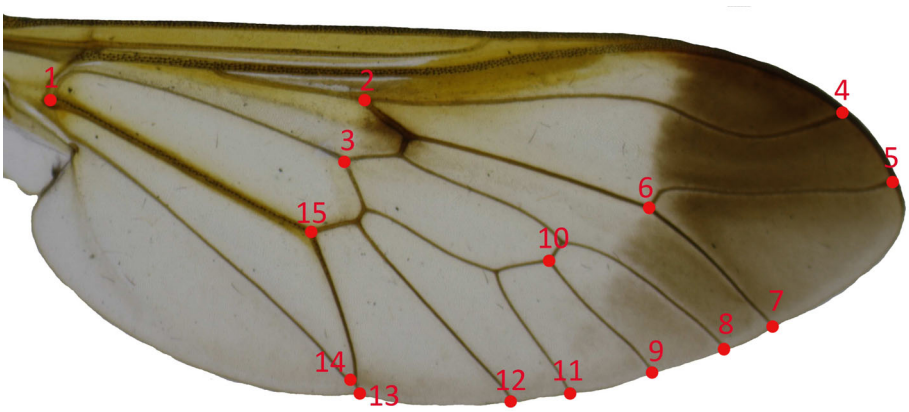

Fig 1 Wing of *Diachlorus jobbinsi* from Caquetá, "PNN Serranía del Chiribiquete" (CAQ), with the configuration of landmarks: *1*=start of second basal cell; *2*=start of veins R2+R3; *3*=start of discal cell; *4*=intersection of the R2+R3 and costal veins; *5*=intersection of R4 and costal veins; *6*=intersection of veins R4 and R5; *7*=end of vein R5; *8*=end of vein M1; *9*=end of vein M2; *10*=end of discal cell; *11*=end of vein M3; *12*=end of vein M4; *13*=end of vein CuA; *14*=intersection of veins CuA and A1; *15*=start of vein CuA.

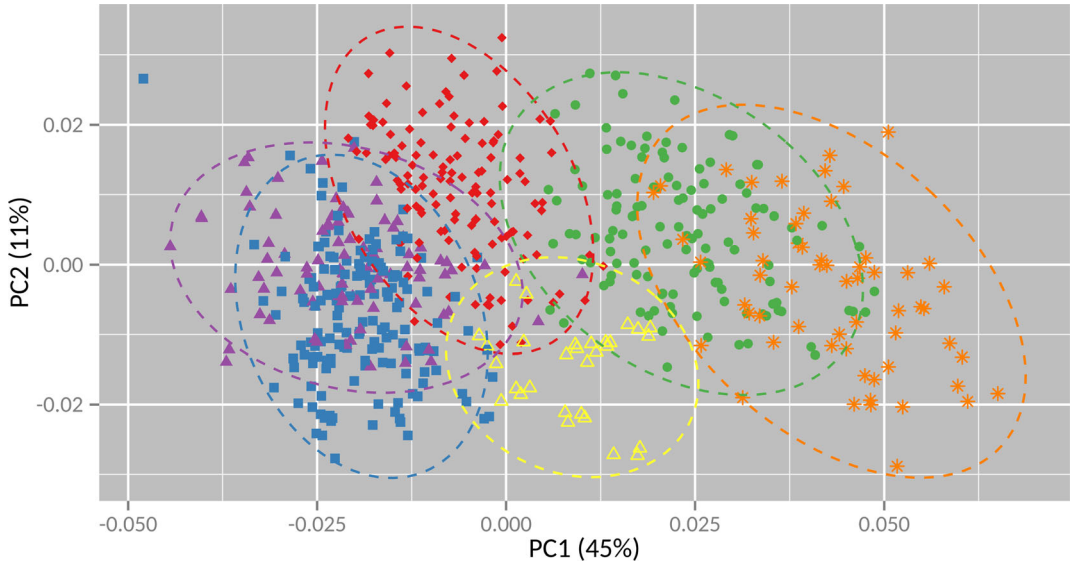

Fig 2 Dispersion graph of genus *Diachlorus* from Colombia in the bidimensional space of principal components PC1 and PC2. The covariance error ellipses (95% of confidence) are drawn. *Red diamonds=Diachlorus curvipes*; *blue squares=Diachlorus fuscistigma*; *green points=Diachlorus jobbinsi*; *yellow empty triangles=Diachlorus leticia*; *violet solid triangles=Diachlorus leucotibialis*; *orange asterisks=Diachlorus nuneztovari*.

A minimal size of specimens was used for each sample ($n>29$) (Table 1), taking into account the within-group variation relative to differences between groups (data not shown), the question and the taxonomic rank addressed, and previous studies that suggest how to evaluate the sample size (Polly 2005; Cardini & Elton 2007). Over each photograph, a configuration of 15 type I landmarks was digitalized (Bookstein 1991) (Fig 1). This configuration was chosen, avoiding type II and III landmarks, to cover the greatest surface of the wing, with the maximum variation, using the fewest number of landmarks.

Morphometric analyses

First, a generalized Procrustes analysis (GPA) was performed for each species, using the least squares criterion (Gower 1975, Rohlf & Slice 1990) over the subsample of the left and right wings to obtain shape variables and consensus shapes. The Riemannian distance was used to quantify the asymmetric shape differences (Kendall 1984, Savriama & Klingenberg 2006). The significance in asymmetry was evaluated for each specimen, through Goodall's *F* test (Goodall 1991, Dryden & Mardia 1998, Savriama & Klingenberg 2006).

Subsequently, a GPA (Gower 1975, Rohlf & Slice 1990) was performed for the entire set of right wings. With the aligned configurations, a principal component analysis (PCA) was performed to reduce the dimensionality of the data and to observe the tendency of the interspecific variation. From the set of shape principal components (PCs) (80% of cumulative explained variance), the following were determined: the relevant number of clusters in the entire data set using the Bayesian information criterion (BIC) for model-based clustering (McLachlan & Peel 2000, Fraley & Raftery 2002, 2007), the "gap" statistic (Tibshirani *et al* 2001), and the Calinski and Harabasz (CH) index (Calinsky & Harabasz 1974).

The unweighted pair group method with arithmetic mean (UPGMA) hierarchical cluster analysis was conducted (Sokal & Sneath 1963, McQuitty 1966, Sneath & Sokal 1973) using Mahalanobis distances computed from the set of principal components. The Riemannian distance was used to quantify the geometric shape differences among the species (Kendall 1984), and Goodall's *F* test calculated the statistical

Table 2 Cumulative percentages (%) of variance explained of the first ten components in PCAs by species and genus.

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9	PC10
Genus *Diachlorus*	45.0	56.0	63.9	70.0	75.5	79.7	83.0	85.7	87.7	89.4
Diachlorus curvipes	23.0	34.7	46.0	55.0	61.0	66.3	71.5	75.4	78.7	82.0
Diachlorus fuscistigma	23.0	36.0	46.3	55.5	62.8	68.5	73.0	76.4	79.7	82.6
Diachlorus jobbinsi	20.0	39.0	52.0	61.7	67.3	72.7	76.4	80.0	83.0	85.7
Diachlorus leucotibialis	24.0	35.4	46.2	55.0	62.3	68.3	73.0	76.8	80.0	83.0
Diachlorus nuneztovari	30.0	44.0	54.0	62.0	69.0	73.8	78.0	81.0	84.0	86.7

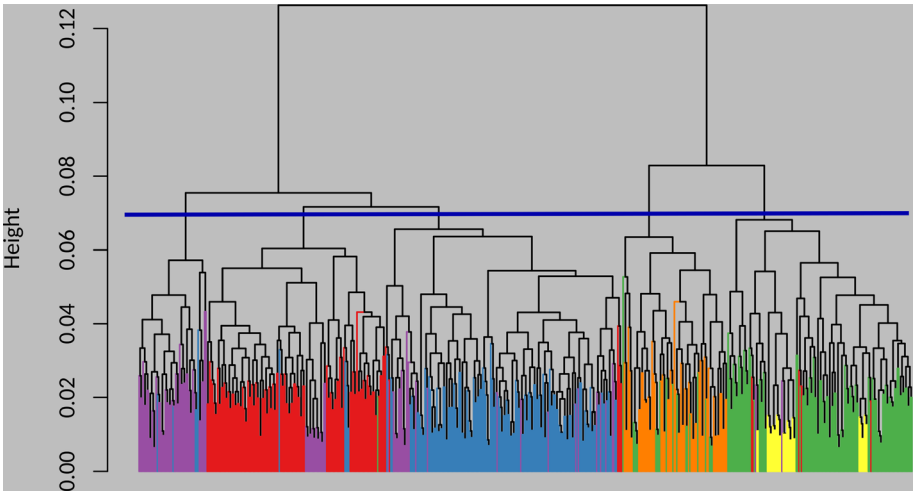

Fig 3 Dendrogram of the genus *Diachlorus* ($K=5$) from Colombia by hierarchical cluster analysis (UPGMA). The *dark blue line* cuts the dendrogram into the *K* groups defined according to the Bayesian information criterion, the "gap" statistic, and the Calinski and Harabasz (CH) index. *Red*=*Diachlorus curvipes*; *blue*=*Diachlorus fuscistigma*; *green*=*Diachlorus jobbinsi*; *yellow*=*Diachlorus leticia*; *violet*=*Diachlorus leucotibialis*; *orange*=*Diachlorus nuneztovari*.

significance of the geometric distances among species (Goodall 1991, Dryden & Mardia 1998). The shape changes among the species were visualized through relative warps: thin-plate spline transformation grids and displacement vectors (Bookstein 1989, 1991, Rohlf & Bookstein 2003).

The same protocol described above for interspecific variation was followed for the measurement of intraspecific variation. The species that were present in more than one area (*D. curvipes*, *D. fuscistigma*, *D. jobbinsi*, *D. leucotibialis*, and *D. nuneztovari*) were used to compare for each species the geometrical wing configurations of specimens from different areas. The Mantel test (Mantel 1967) was used to evaluate the correlation between the matrix of Riemannian distances and the matrix of geographical distances. The analyses were performed separately for each species, and the unities of differentiation were the areas where the species are distributed.

Finally, with the calculated Mahalanobis distances from the shape PCA, the discriminant analysis (Fisher 1936, Slice 2007) (linear and quadratic) was carried out to predict both, species membership and area (locality) membership. Cross-validation was used to obtain the posterior classification probabilities in which each individual is allocated to its closest group without being used to determine the centroid of the group (Venables & Ripley 2002, Tuffery 2011).

The package "geomorph" was used for the gathering of landmark data (Adams & Otarola-Castillo 2013) and the packages "cluster" (Maechler *et al* 2014), "DiscriMiner" (Sanchez 2013), "ellipse" (Murdoch & Chow 2013), geomorph (Adams & Otarola-Castillo 2013), "mclust" (Fraley & Raftery 2002), "NbClust" (Charrad *et al* 2014), "shapes" (Dryden 2013), and "vegan" (Oksanen *et al* 2013) for the described statistical analysis. The packages "colorspace" (Ihaka *et al* 2013) and "ggplot2" (Wickham 2009) were used for graphics. All packages were from the R software environment for statistical computing and graphics (R Development Core Team 2014). The morphometric and statistical analyses can be reproduced using the code and data available in the repository https://github.com/atorresgalvis/Diachlorus_Colombia-Wing_shape.

Table 3 Riemannian distances among *Diachlorus* species from Colombia, based on the wing shape.

	Diachlorus curvipes	*Diachlorus fuscistigma*	*Diachlorus jobbinsi*	*Diachlorus leucotibialis*	*Diachlorus leticia*
Diachlorus fuscistigma	0.02587*				
Diachlorus jobbinsi	0.03292*	0.04486*			
Diachlorus leucotibialis	0.02566*	*0.01807**[a]	0.04619*		
Diachlorus leticia	0.03685*	0.03366*	0.02783*	0.03925*	
Diachlorus nuneztovari	0.05359*	0.06362*	0.02695*	*0.06568**[b]	0.04453*

Italicized values are the minimum (superscript a) and maximum (superscript b) Riemannian distances.

*$p<0.05$, in Goodall's *F* test.

Table 4 Riemannian distances among groups of specimens from different areas.

Diachlorus curvipes				
	AMA	CHO	CHO	
CHO	*0.01239**[b]			
MET	0.01200*	0.00872*		
PUT	*0.00594*[a]	0.01128*	0.01159*	
Diachlorus fuscistigma				
	AMA	CAQ	CAQ	PUT
CAQ	0.00894*			
MET	*0.01321**[b]	0.01235*		
PUT	*0.00816*[a]	0.01095*	0.01276*	
VAU	0.00894*	0.00820	0.01235*	0.01095*
Diachlorus jobbinsi				
	AMA	CAQ	CHO	
CAQ	0.01189*			
CHO	*0.02497**[b]	0.02138*		
PUT	*0.01135**[a]	0.01660*	0.02426*	
Diachlorus leucotibialis				
	AMA	CAQ		
CAQ	*0.01977**[b]			
PUT	*0.01077**[a]	0.01587*		

Italicized values are the minimum (superscript a) and maximum (superscript b) Riemannian distances. For *Diachlorus nuneztovari*, not present in the table, the Riemannian distance between AMA and PUT is 0.01596 (asterisk).

AMA Amazonas, *CAQ* Caquetá, *CHO* Chocó, *MET* Meta, *PUT* Putumayo, *VAU* Vaupés.

*$p<0.05$, in Goodall's *F* test.

Results

Wing asymmetry

There was no significant variation in left/right wings; therefore, any of them could be used in geometric morphometrics studies. The mean Riemannian distances between left and right wings ranged from 0.003413 (*D. leucotibialis*, $G=0.297$, $p=0.99$) to 0.005972 (*D. curvipes*, $G=0.785$, $p=0.63$).

Interspecific wing variation

The ordination of the specimens in the morphospace defined by the principal components 1 (PC1) and 2 (PC2) (cumulative variance=56%) could differentiate at least four of the six species of *Diachlorus* in the study (Fig 2). However, it was necessary to use the first seven principal components to retain 80% of the cumulative explained variance (Table 2).

The optimal number of clusters (*K*) for the wing geometry data was 5, based on the BIC, the gap statistic, and the CH index, with an ellipsoidal, equal volume, shape, and orientation ("EEE") Gaussian mixture model (see the Electronic Supplementary Material). These five *K* groups almost perfectly represented the species *D. curvipes*, *D. fuscistigma*, *D. jobbinsi*, *D. leucotibialis*, and *D. nuneztovari* in the hierarchical cluster analysis (Fig 3). Specimens of *D. leticia* were clustered together, though nested within *D. jobbinsi* (Fig 3), meaning that the cluster of *D. jobbinsi* was, in fact, *D. jobbinsi*+*D. leticia*.

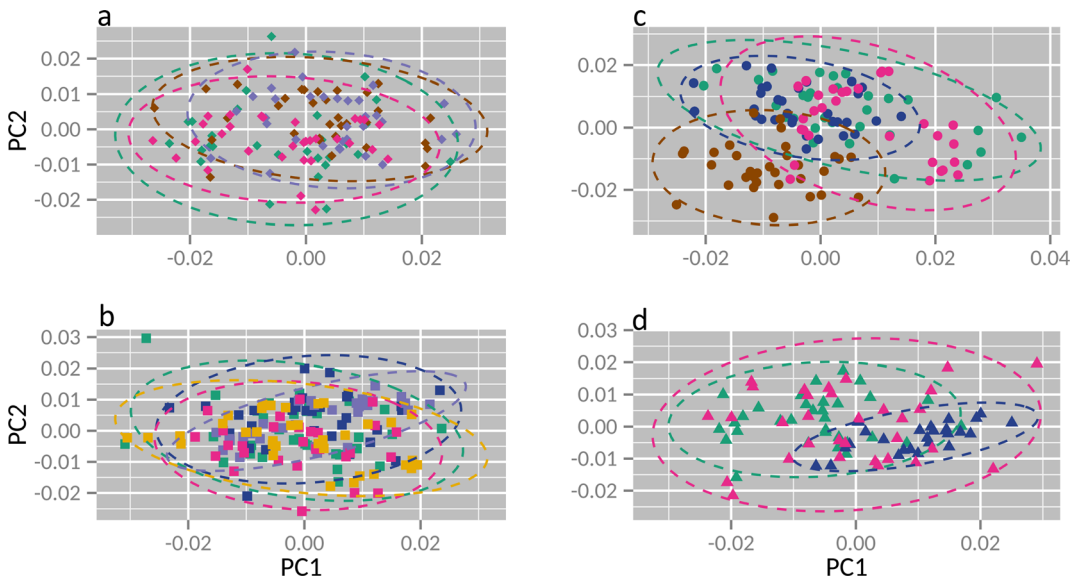

Fig 4 Dispersion graphs of four *Diachlorus* species from different areas of Colombia in the bidimensional space of principal components PC1 and PC2. The covariance error ellipses (95% of confidence) are drawn. *a* (*diamonds*)=*Diachlorus curvipes* (PC1=23%, PC2=11.7%); *b* (*squares*)=*Diachlorus fuscistigma* (PC1=23%, PC2=13%); *c* (*points*)=*Diachlorus jobbinsi* (PC1=20%, PC2=19%); *d* (*triangles*)=*Diachlorus leucotibialis* (PC1=24%, PC2=11.4%). *Aquamarine green*=Amazonas; *dark blue*=Caquetá; *brown*=Chocó; *light purple*=Meta; *pink*=Putumayo; *dark yellow*=Vaupés (see the Electronic Supplementary Material for the PCA of *Diachlorus nuneztovari*).

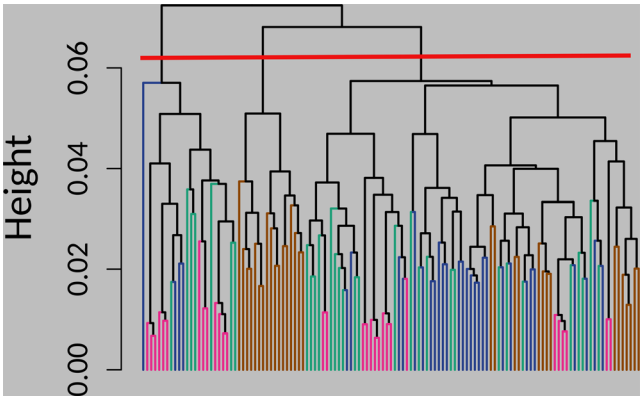

Fig 5 Dendrogram of *Diachlorus jobbinsi* (*K*=3) from different areas of Colombia by hierarchical cluster analysis (*UPGMA*). The *red line* cuts the dendrogram into the *K* groups defined according to the Bayesian information criterion, the "gap" statistic, and the Calinski and Harabasz (*CH*) index. *Aquamarine green*=Amazonas; *dark blue*=Caquetá; *brown*=Chocó; *pink*=Putumayo.

The differences among all species were statistically significant ($p<0.05$ for all species). *Diachlorus nuneztovari* was the species geometrically most distant from the other species, while *D. curvipes* was the least distant from the others (Table 3). The most similar pair of species was *D. fuscistigma* and *D. leucotibialis*, while the most different pair was *D. leucotibialis* and *D. nuneztovari* (Table 3).

Intraspecific wing variation

Goodall's *F* test showed significant shape differences ($p<0.05$) among groups of specimens from different areas in all species, except between Amazonas and Putumayo for the species *D. curvipes* and *D. fuscistigma* and between Caquetá and Vaupés for *D. fuscistigma* (Table 4). Riemannian distances (Table 4) showed that, for *D. curvipes*, the specimens from Chocó were the most different in relation with the specimens from the other areas while specimens from Putumayo were the least different; for *D. fuscistigma*, Meta and Vaupés were the most and least different, respectively; in *D. jobbinsi*, Chocó was the most different and Caquetá was the least different; and for *D. leucotibialis*, Caquetá was the most different and Putumayo was the least different. The Mantel test denoted that the Riemannian distances were not significantly correlated with geographical distances between the areas; *D. jobbinsi* had the highest correlation (r^2= 0.3460, p=0.217), while *D. leucotibialis* had the lowest (r^2=0.2331, p=0.667).

The cumulative explained variance of the first two components (PC) for all species was low (38% on average), such that, for example, in *D. curvipes*, it was necessary to use the first ten components to retain 80% of the explained variance (Table 2). In *Diachlorus*, it was not possible to determine the geographic origin of the specimens, based on the morphospace defined by the PC1 and PC2 of the wing shape (only one *K* group was determined by BIC, the gap statistic, and the CH index with a diagonal multivariate normal "XXI" model; for each species, see the Electronic Supplementary Material). The only exception was *D. jobbinsi*, where the specimens from Chocó were well separated from the other parks (Fig 4); the optimal number

Table 5 Confusion matrix of cross-validated classification of the *Diachlorus* species.

	F[ua]	*Diachlorus curvipes*	*Diachlorus fuscistigma*	*Diachlorus jobbinsi*	*Diachlorus leucotibialis*	*Diachlorus leticia*	*Diachlorus nuneztovari*
Diachlorus curvipes	Lin	*97.6*		1.60	0.80		
	Qua	*98.4*		1.60			
Diachlorus fuscistigma	Lin	0.70	*96.8*		2.50		
	Qua		*96.8*		3.10		
Diachlorus jobbinsi	Lin			*85.6*			14.4
	Qua			*89.6*			10.4
Diachlorus leucotibialis	Lin	1.10	7.70		*90.1*	1.10	
	Qua	1.10	7.70		*90.1*	1.10	
Diachlorus leticia	Lin					*100*	
	Qua					*100*	
Diachlorus nuneztovari	Lin			3.30			*96.7*
	Qua			5.00			*95.0*

The percentages of assignments to the species are presented. Italicized values are the correct assignments. The misclassified global error rate is 0.101 for linear function and 0.406 for quadratic function.

Lin linear, *Qua* quadratic.

[a] Discriminant functions are linear and quadratic.

of clusters (*K*) was 3, with a diagonal, equal volume and shape ("EEI") model. The hierarchical clustering of *D. jobbinsi* showed that more than a half of the specimens from Chocó were grouped together into one *K* group but the specimens from the other parks were mixed into the other two groups (Fig 5).

Shape variation and taxonomic discrimination

The linear discriminant analysis was better than the quadratic in predicting species membership, while the quadratic discriminant analysis overcame the linearity in predicting area membership. *Diachlorus leticia* was the best-assigned species, reaching 100% of accuracy, while *D. jobbinsi* (89.6%) had the lowest accuracy. The accuracy was lower with a quadratic discriminant in *D. nuneztovari* (Table 5). The specimens of *D. jobbinsi* and *D. leucotibialis* were the best assigned to the areas from where they came, while *D. fuscistigma* presented the worst assignment (Table 6).

Visualization of shape changes

The intraspecific and interspecific changes were not the same along the wing, and these differences regarding variation were represented not only in the amount of change, but the morphological localization on the wing. The diagrams of interspecific variation (Fig 6a, b) showed large displacements upward and backward on the apex of the wing (intersections of the radial and costal veins; apical and costal wing border), downward and forward on the posterior/anal wing section (ends of veins M3, M4, and CuA; intersection of veins CuA and A1), and backward on the intersection of veins R4 and R5. In intraspecific deformations (Fig 6c, d), as expected, the diagram of minimum variation showed almost no changes, while the diagram of maximum variation showed displacements forward on the apex of the wing, and changes on the separation distance between the intersection of veins CuA and A1 and the posterior wing border.

Discussion

Geometric morphometrics could agree with the results from traditional external morphology. *Diachlorus nuneztovari* was similar to *D. jobbinsi* (Fig 2, Table 3), and this result agreed with the analysis of external morphology that placed these two species very close, mainly in terms of coloration of the thorax and legs, making their separation in some occasions using external morphology more difficult (Fairchild 1972, Wilkerson & Fairchild 1982).

Table 6 Cross-validated classification of the areas where four *Diachlorus* species are distributed.

Diachlorus curvipes						
	F[ua]	AMA	CHO	MET	PUT	
AMA	Lin	*51.8*	6.80	10.3	31.1	
	Qua	*75.6*		10.3	13.8	
CHO	Lin	3.10	*62.6*	21.8	12.5	
	Qua	3.10	*84.5*	9.30	3.10	
MET	Lin	9.30	12.5	*71.9*	6.30	
	Qua	3.10	9.30	*84.4*	3.20	
PUT	Lin	18.7	12.5	15.6	*53.2*	
	Qua	18.7	12.5		*68.8*	
Diachlorus fuscistigma						
	F[ua]	AMA	CAQ	MET	PUT	VAU
AMA	Lin	*50.0*	9.30	3.10	12.6	25.0
	Qua	*65.7*	9.30		6.20	18.8
CAQ	Lin	15.6	*53.2*	9.30	6.30	15.6
	Qua	6.20	*78.1*	3.10	6.30	6.30
MET	Lin			*90.6*		9.40
	Qua			*100*		
PUT	Lin	21.4	21.4	10.8	*25.0*	21.4
	Qua	14.3	14.3	3.50	*53.6*	14.3
VAU	Lin	28.6	22.8	2.80	14.2	*31.6*
	Qua	11.4	11.4		8.60	*68.6*
Diachlorus jobbinsi						
	F[ua]	AMA	CAQ	CHO	PUT	
AMA	Lin	*64.5*	19.3		16.2	
	Qua	*83.9*	9.70		6.40	
CAQ	Lin	6.30	*84.4*		9.30	
	Qua	18.8	*81.2*			
CHO	Lin		3.20	*96.8*		
	Qua			*100*		
PUT	Lin	16.7			*83.3*	
	Qua				*100*	
Diachlorus leucotibialis						
	F[ua]	AMA	CAQ	PUT		
AMA	Lin	*78.2*		21.8		
	Qua	*100*				
CAQ	Lin		*96.3*	3.70		
	Qua		*100*			
PUT	Lin	6.30	12.5	*81.2*		
	Qua	3.10		*96.9*		

The percentages of assignments to areas are presented. Italicized values are the correct assignments. The misclassified global error rate for discriminant functions are as follows: *Diachlorus curvipes* (Lin=0.648; Qua=0.472), *Diachlorus fuscistigma* (Lin=0.660; Qua=0.553), *Diachlorus jobbinsi* (Lin=0.288; Qua=0.224), and *Diachlorus leucotibialis* (Lin=0.439; Qua=0.230). (*Diachlorus nuneztovari* confusion matrix is in the Electronic Supplementary Material).

AMA Amazonas, *CAQ* Caquetá, *CHO* Chocó, *MET* Meta, *PUT* Putumayo, *VAU* Vaupés, *Lin* linear, *Qua* quadratic.

[a] Discriminant functions are linear and quadratic.

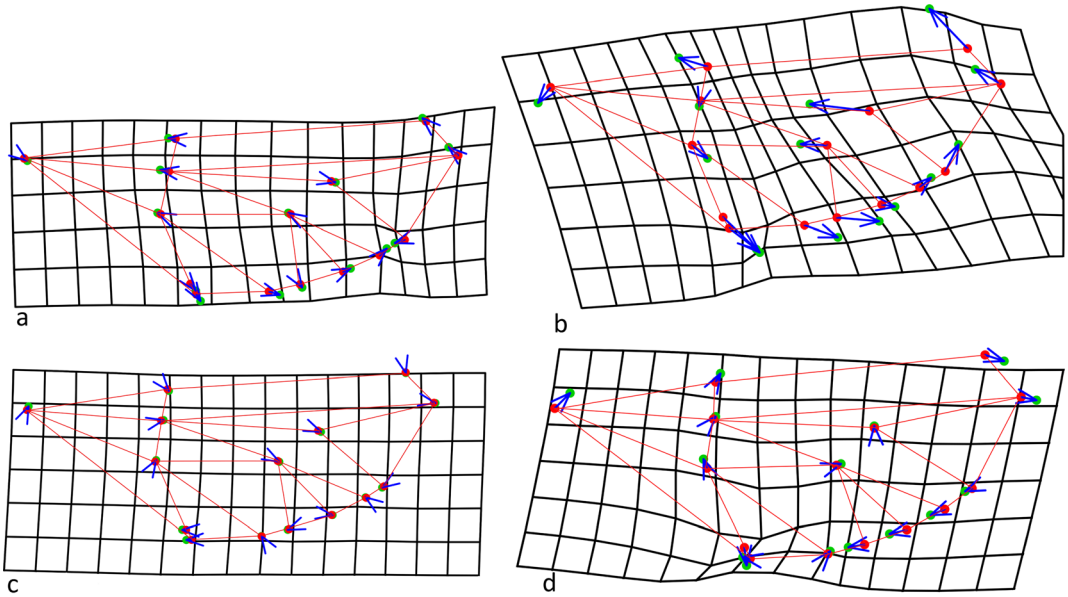

Fig 6 Deformation diagrams, using thin-plate spline grids over consensus shapes. **a** From *Diachlorus leucotibialis* to *Diachlorus fuscistigma* (minimum interspecific variation). **b** From *Diachlorus leucotibialis* to *Diachlorus nuneztovari* (maximum interspecific variation). **c** *Diachlorus fuscistigma* from Vaupés to Putumayo (minimum intraspecific variation). **d** *Diachlorus jobbinsi* from Amazonas to Chocó (maximum intraspecific variation). *Red points* are landmark configurations of source consensus shapes, while *green points* are landmark configurations of target consensus shapes; *blue arrows* (or *blue arrowheads*) indicate directions and amount of shape changes from source to target consensus shapes.

Other geometrical analyses could disagree with traditional morphology. *Diachlorus fuscistigma* and *D. leucotibialis* were quite similar using morphometrical data, contrasting with the analysis of external morphology that allows to clearly differentiate these species based on coloration characters of the thorax and abdomen (Wilkerson & Fairchild 1982). *Diachlorus curvipes* and *D. leticia* species could be distinguished very well from the rest of the species under study (Fig 3).

Geometric morphometrics allowed to define the origin of the specimen for a given species. Chocó, the farthest area geographically from the others in this study, presented the specimens that were the most different in wing shape. This can be seen in the two species presented in Chocó, *D. jobbinsi* and *D. curvipes* (Table 4), in which the specimens from Chocó have the most different wing geometry among the areas where these species are present. However, there was a nonsignificant correlation between Riemannian and geographical distances.

This finding disagreed with those obtained in other groups of Diptera for many morphological structures (Dujardin *et al* 1999, Franco *et al* 2008), which could indicate that morphological intraspecific variation, wing shape of *Diachlorus*, in this case, could be related to other variables not taken into account in this study (such as environmental and ecological conditions, genetic variability, and adaptive variables) and not only to the geographic proximity (Lane & Marshall 1981, Soto *et al* 2007, Franco *et al* 2008). Besides, the groups formed in the cluster analysis for *D. jobbinsi* did not always correspond to predefined groups of specimens from each area (Figs 4 and 5). This means that intraspecific variation is due not only to geographic distribution but also to different variables that can be genetic or ecological (Lane & Marshall 1981, Bolnick *et al* 2011). However, the smaller-ranged species have low intraspecific morphological variation and tend to occupy smaller niches (Oney *et al* 2013). This pattern is not supported in the present study where the widest-ranged species, *D. curvipes* and *D. fuscistigma*, have the lowest intraspecific variation while smaller-ranged species, *D. jobbinsi*, *D. leucotibialis*, and *D. nuneztovari*, have the highest intraspecific variation (Fig 4, Table 4).

The interspecific variation in *Diachlorus* required fewer principal components and variables to describe its structure, and it fits very well to a linear discriminant function, while intraspecific variation required the addition of more principal components and fits better to a quadratic discriminant function (Tables 5 and 6). Linear discriminant analysis is conceptually simple and has been used broadly (Dujardin *et al* 1999, De la Riva *et al* 2001, Sheets *et al* 2006, Jaramillo *et al* 2014). However, it has some limitations: it requires at least one of the scatter matrices to be nonsingular, it cannot easily capture a nonlinearly clustered structure, and it allows low-dimensional representations of the data (Roth & Steinhage 1999).

Interspecific variation could be interpreted as cumulative intraspecific changes (Solé *et al* 1999, Jablonski 2000, Simons 2002), which means that mechanisms of intraspecific and interspecific variations are the same. Then, the only difference between intraspecific and interspecific variations would

be the amount of change that would tend to be higher in interspecific variation than in intraspecific variation. We did not observe this pattern in *Diachlorus*; its intraspecific and interspecific mechanisms of wing variation are different not only in places of the wing but also in direction and amount of variation (Fig 6).

The effectiveness of determining the species and the area of origin in the genus *Diachlorus* can be viewed as an inverse behavior from the amount of intraspecific variation to the amount of interspecific variation, because in those species with more accurate predicting species membership, predicting area membership is less accurate (*D. curvipes*, *D. fuscistigma*); however, species that are more accurate in predicting area membership are less accurate in predicting species membership (*D. jobbinsi*, *D. leucotibialis*) (Tables 5 and 6).

Geometric morphometrics is a powerful complement to the external morphology in the taxonomy of *Diachlorus*, and given that geometric morphometrics techniques are efficient in the taxonomic understanding of *Diachlorus*, we hypothesize that the rest of the species of *Diachlorus* that are not in Colombia can also be taxonomically discriminated using geometric morphometrics of the wings. The expansion of the range of data collection could define more accurately the dynamics of regional variation. For these reasons, further morphometric studies of populations (not only of *Diachlorus* but also of other genera of Tabanidae) along the entire distribution range would help to evaluate in a better way morphological variation in Tabanidae.

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